

SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				kills bacteria / microbes	1			1		
2				electronegativity increases across Period 3	1			1		
3				phosphorus	1			1		
4				3s 3p 3d 4s ↑↓ ↑↓↑↓↑↓ ↑↑↑ ↑↓	1			1		
5				2 Mg(NO ₃) ₂ → 2 MgO + 4 NO ₂ + O ₂		1		1		
6	(a)			disagree – since 16 neutrons in ³² ₁₆ S ²⁻ and 10 neutrons in ¹⁸ ₈ O			1	1		
	(b)			agree – since 18 electrons in both species			1	1		
7				4.5		1		1		
8				22.4 dm ³ = 1 mol $n = \frac{1}{22.4} \quad (1)$ 1 dm ³ = 0.0446 mol $M_r = \frac{0.717}{0.0446} = 16.1 \quad (1)$		2		2	1	
				Section A total	4	4	2	10	1	0